Cora [23] and MNIST [19]. The experimental results demonstrate that our focal diversity based hierarchical pruning approach is effective in identifying high quality deep ensembles with significantly smaller sizes and better ensemble accuracy than the entire deep ensemble team.

2 ENSEMBLE PRUNING WITH FOCAL DIVERSITY

Given an entire deep ensemble with a large size M, it consists of M individual member networks ($F_i, i \in \{0, 1, ..., M-1\}$) that are trained for a specific learning task and dataset. We denote the set of all possible sub-ensembles as $EnsSet$, which are composed of any subset of these M individual networks. For a specific team size S, let $EnsSet(S)$ denote the set of all possible sub-ensembles of size S in $EnsSet$. The cardinality of $EnsSet(S)$ is calculated based on the selection of S networks from all M base networks, that is $|EnsSet(S)| = \binom{M}{S}$. Therefore, the total number of candidate sub-ensembles for $S = 2, \ldots, M-1$ is $|EnsSet| = \sum_{S=2}^{M-1} |EnsSet(S)| = \binom{M}{2} + \binom{M}{3} + \ldots + \binom{M}{M-1} = 2^M - (2+M)$, which grows exponentially with M. For example, when $M = 3,\ 5,\ 10,\ 20$, we have $|EnsSet| = 3,\ 25,\ 1012,\ 1048554$ respectively. It may not be feasible to perform the exhaustive search of all possible ensembles in $EnsSet$ with a large M. Hence, it is critical to develop efficient ensemble pruning methods for examining the sub-ensembles from the candidate set $EnsSet$, removing these sub-ensembles with low diversity and obtaining the set $GEnsSet$ of high quality sub-ensembles with lower cost and better ensemble accuracy than the entire deep ensemble of M member networks.

Table 3. Example Deep Ensembles for CIFAR-10 and ImageNet

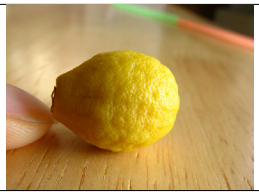
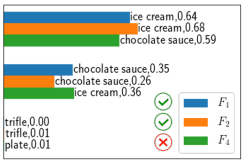
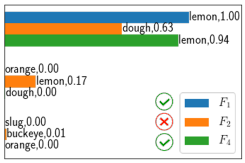
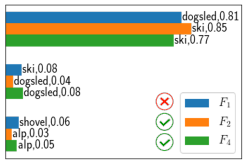
Dataset	CIFAR-10					ImageNet				
Ensemble Team	0123456789	0123	01238	123	1234	0123456789	12345	2345	1234	124
Ensemble Acc (%)	96.33	**97.15**	**96.87**	**96.81**	**96.63**	79.82	**80.77**	**80.70**	**80.29**	**79.84**
Team size	10	4	5	3	4	10	5	4	4	3
Acc Improv (%)	0	**0.82**	**0.54**	**0.48**	**0.30**	0	**0.95**	**0.88**	**0.47**	**0.02**
Cost Reduction (%)	0	**60**	**50**	**70**	**60**	0	**50**	**60**	**60**	**70**

Table 4. All Individual Member Models for Four Benchmark Datasets

Dataset	CIFAR-10		ImageNet		Cora		MNIST	
	10,000 testing samples		50,000 testing samples		1,000 testing samples		10,000 testing samples	
ID	Name	Acc (%)	Name	Acc (%)	Name	Acc (%)	Name	Acc (%)
0	**DenseNet190**	**96.68**	AlexNet	56.63	GCN	81.70	KNN	94.23
1	DenseNet100	95.46	DenseNet	77.15	GAT	82.80	Logistic Regression	91.89
2	ResNeXt	96.23	EfficientNet-B0	75.80	SGC	81.70	Linear SVM	92.48
3	WRN	96.21	ResNeXt50	77.40	ARMA	82.10	**RBF SVM**	**96.31**
4	VGG19	93.34	Inception3	77.25	APPNP	82.20	Random Forest	95.91
5	ResNet20	91.73	**ResNet152**	**78.25**	APPNP1	83.80	GBDT	92.89
6	ResNet32	92.63	ResNet18	69.64	APPNP2	88.70	Neural Network	96.18
7	ResNet44	93.10	SqueezeNet	58.00	**SplineCNN**	**88.90**		
8	ResNet56	93.39	VGG16	71.63	SplineCNN1	88.30		
9	ResNet110	93.68	VGG19-BN	74.22	SplineCNN2	88.50		
MIN	ResNet20	91.73	AlexNet	56.63	GCN/SGC	81.70	Logistic Regression	91.89
AVG		94.25		71.60		84.87		94.27
MAX	DenseNet190	96.68	ResNet152	78.25	SplineCNN	88.90	RBF SVM	96.31

We show 5 example deep ensemble teams in Table 3 for two datasets, CIFAR-10 and ImageNet. For each dataset, we list the entire deep ensemble (`0123456789`) of 10 member networks in addition to the 4 high quality deep ensembles that are identified by our HQ ensemble pruning approach. The 10 member networks in each entire ensemble for CIFAR-10 and ImageNet are given in Table 4. The

Table 5. Examples on ImageNet and Top-3 Classification Confidence

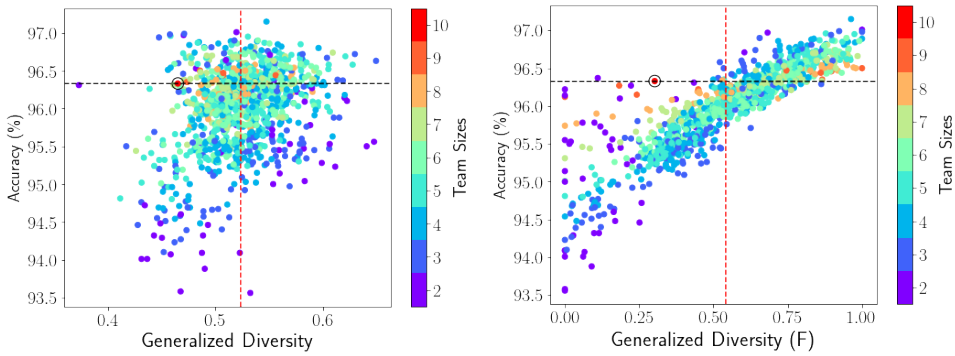
Image			
Ground Truth Label	ice cream	lemon	ski
Accuracy (%) $F_1F_2F_4$: 79.84 (DenseNet: 77.15, EfficientNet-B0: 75.80, Inception3: 77.25)			
Ensemble Output	ice cream	lemon	ski

4 sub-ensembles recommended by our HQ have much smaller team sizes with only 3∼5 individual member networks and achieve better ensemble accuracy than the given entire ensemble of 10 networks, significantly reducing the ensemble execution cost by 50%∼70% for both CIFAR-10 and ImageNet. Table 5 further presents 3 image examples from ImageNet and the prediction results with Top-3 classification confidence from the member networks of the sub-ensemble team 124. This ensemble team achieves higher accuracy than each individual member network. For each image, one member model makes a prediction error. But the ensemble team can still give the correct predictions by repairing the wrong predictions by its member networks. This sub-ensemble team of 3 member networks also outperforms the entire ensemble of 10 models on ImageNet. For a given deep ensemble of size $M = 10$, we have a total of 1012 possible sub-ensembles to be considered in ensemble pruning. It is challenging to design and develop an efficient ensemble pruning approach to identify high quality sub-ensembles to improve both ensemble accuracy and time and space efficiency of ensemble execution.

Problems with Baseline Ensemble Pruning. In related work, we discussed several recent studies that utilize deep ensembles to strengthen the robustness of individual deep neural network against adversarial attacks [8, 9, 20, 34]. Most of these existing methods leverage Cohen's Kappa (CK) [25] for measuring ensemble diversity since early studies [17, 24, 31, 32] in the literature have shown that both pairwise and non-pairwise diversity metrics share similar diversity evaluation results with regard to ensemble predictive performance, including the CK, BD, KW and GD metrics mentioned in related work. However, we show that these existing diversity metrics may not precisely capture the inherent failure independence among member networks of a deep ensemble team, which may not produce the optimal performance in guiding ensemble pruning.

We first study the baseline diversity metrics for ensemble pruning and analyze their inherent problems. For a given diversity metric, such as BD or GD, the baseline ensemble pruning method calculates the ensemble diversity scores for each candidate ensemble in *EnsSet* using a set of random samples drawn from the validation set as suggested by [17]. A practical method for pruning a given large ensemble of M member networks follows three steps: (1) we first compute the ensemble diversity score for each candidate sub-ensemble by using CK, BD, KW or GD; (2) we calculate the mean diversity score as the pruning threshold; and (3) we choose these sub-ensembles in *EnsSet* with their ensemble diversity scores above this pruning threshold and add them into the set

GEnsSet of high quality ensembles. The rest sub-ensembles will be pruned out given their lower ensemble diversity than the pruning threshold.



(a) **GD**, Pruning 427 out of 1012 (b) **F-GD**, Pruning 517 out of 1012

Fig. 1. Pruning All Possible Deep Ensembles by The Mean Threshold on CIFAR-10: (a) Baseline Diversity (GD), (b) Focal Diversity (F-GD)

We compute the baseline GD scores and ensemble accuracy for all 1013 (1012 sub-ensembles + 1 entire ensemble) deep ensembles on CIFAR-10 as shown in Figure 1a, where each dot denotes an ensemble team and each color marks the team size in the right color mapping bar. The baseline GD ensemble pruning approach utilizes the GD mean threshold of 0.524 (vertical red dashed line) to select sub-ensemble teams on the right of this diversity threshold line into *GEnsSet*. The ensemble accuracy 96.33% of the entire ensemble (marked by the black circle) is represented by the horizontal black dashed line, which serves as the reference accuracy for evaluating ensemble pruning methods. Ideally, the ensemble pruning algorithms should identify high quality sub-ensembles with their ensemble accuracy above this reference accuracy.

For an entire ensemble of a large size M, we use four performance metrics to assess the ensemble pruning algorithms: (1) the accuracy range of the selected sub-ensembles in *GEnsSet*, (2) the precision, which measures the proportion of the selected sub-ensembles with equal or higher ensemble accuracy than the reference accuracy of the given entire ensemble, e.g., 96.33% accuracy for CIFAR-10 and 79.82% accuracy for ImageNet, among all selected sub-ensembles, (3) the recall, which measures the proportion of the selected sub-ensembles in *GEnsSet* over all the sub-ensembles in *EnsSet* whose ensemble accuracy is equal to or higher than the reference accuracy of the entire deep ensemble, and (4) the cost reduction, which measures the reduction in the ensemble team size of the selected sub-ensembles relative to the entire ensemble size M. For example, we show a small sub-ensemble, 123, in Table 3 with the cost reduction of 70% ($= (M - S)/M = (10 - 3)/10$). We evaluate the baseline mean threshold based ensemble pruning algorithm using the baseline BD (2nd column) and GD (4th column) diversity metrics on the CIFAR-10 dataset in Table 6 and ImageNet dataset in Table 7. The BD-based baseline pruning suffers from low precision (6.18% for CIFAR-10 and 8.53% for ImageNet) and recall (11.72% for CIFAR-10 and 23.15% for ImageNet). Even though the GD-based baseline pruning has higher precision (36.90% for CIFAR-10 and 17.74% for ImageNet) and recall (63.10% for CIFAR-10 and 46.31% for ImageNet), which is still below the acceptable values, such as precision of over 50%. Moreover, the accuracy lower bound for both BD and GD based baseline ensemble pruning, i.e., 93.56% on CIFAR-10 and 61.39% (BD) and 70.79% (GD) on ImageNet, is much lower than the reference accuracy of the entire ensemble team, i.e.,